PROJECTS & ASSIGNMENTS PORTFOLIO

Brian Colquhoun, Hugh O'Donnell and Partners (BCHOD) have successfully worked on several assignments in Zambia and elsewhere in the Southern Africa region since 1948. The efforts of Brian Colquhoun, Hugh O'Donnell and Partners have contributed to the success of several infrastructure projects in Zambia. These include:

TRANSPORTATION & ROADS

- Lusaka Decongestion Project: 2017
- Rehabilitation of the Kalulushi to Lufwanyama (M18) road.
- Upgrading to Bituminous standard and rehabilitation of selected urban roads in Lusaka (Lot 7 & 9) formula 1 road.
- Design and Construction Supervision of the Mazyopa Trash Trap Access road under the Millennium Challenge Account (MCA) project – 2016
- Design and Construction supervision of the ZARI Mount Makulu roads - 2016
- Road Condition surveys and Bridge Infrastructure Surveys of the entire Road and Bridge network in Mozambique – 2010 - 2012
- Consultancy Services for the Design and construction/upgrading of selected Chalala Roads, Mosi-o-Tunya Road Extension, part of Shantumbu Road and South of Chilenje South newly constructed Road Parallel to ZESCO Pylons based on Integrated Construction Unit (ICU) method of works.
- Consultancy Services for the construction supervision of Road T004 from the Airport Turnoff to Luangwa Bridge in Lusaka Province
- Supervision of the Periodic Maintenance of D020 Kapatu to Kasama (from Kandondo to Tamina) at the Junction to Luwingu and Kasama) in Northern Province (72.4Km)
- Accelerated Urban Roads Rehabilitation Programme 2007 - Solwezi
- Accelerated Urban Roads Rehabilitation Programme Rehabilitation of Urban Roads in Mongu
- Consultancy Services for Supervision of Periodic Maintenance works of T005 from Solwezi via Mutanda-Mwinilunga in North-Western Province
- Consulting Services for the Supervision of the Rehabilitation of Road T2 between Lusaka and Chirundu
- Accelerated Urban Roads Rehabilitation Programme in Livingstone
- Supervision of Periodic Maintenance of Stretches Between Km 0 and Km 36, Km 102 and Km 168 of the Road T5 between Chingola and Solwezi

Accelerated urban roads rehabilitation programme 2006/2007 – Mongu Town.

Mongu Accelerated Urban Roads Project was carried out on a Design and Build basis. There were six selected roads namely President Avenue, Airport Road, Libonda road, Arthur Wina Road, Kambule Street and Hospital Road, with total length of 8.13km and 2,100sm car park, of the 8.13 Km, 3 km was rehabilitation and 5.13Km was full construction. The Project value was US $1.4million

Arthur Wina Road before

The main works included the civil engineering design, supervision and construction of bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel wearing course and replenishing or reshaping shoulders and roadside drainage.

Field investigation included Visual survey, traffic survey, topography survey and material survey of the existing pavement and prospective borrow pits.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit.

Arthur Wina Road, Laying of the Second seal (Slurry)

On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal.

Pavement Assessment using the FWD

Rehabilitation of Mukobeko Road and other township roads were carried out under the periodic maintenance of T2 Lusaka- Kabwe Road funded by European Union.

Mukobeko Road was in a very poor condition. It was badly potholed and had lost surface dressing on the last approximately 4.0Km section.

Assessment of the existing Pavement was carried using the Falling weight Deflectometer (FWD) and the Dynamic cone Penetrometer (DCP). Manual traffic counts were carried out for seven days and an economic study was carried out to project the future expected traffic loading

Paver, Laying of 30mm thick Hot Premix Concrete Asphalt.

Mixing of 3% Cement to Gravel before compacting to a depth of 150mm thick as base.

The consultant produced Drawings, Bill of Quantities, Cost Estimates, and Technical Specifications.

The Interventions recommended were to scarifying the existing base, importing some material, mixing and compacting as subbase and then import additional material and lay a new 3% cement stabilised base.

Hot Premix Asphalt on the dual carriageway

The surface received 30mm thick Hot Premix Asphalt concrete.

The Project included rehabilitating the dual carriageway passing through the main town centre and the connected roads. The road markings were also done to define the movement order of the vehicles on different junctions.

Accelerated Urban Roads Rehabilitation Programme 2004/2005 – Lusaka City

Lusaka Accelerated Urban Roads Project was carried out on a Design and Build basis. There were fourteen (14no.) selected roads namely Kabelenga, Tuleteka, Longolongo, Sadzu, seventh street, Reedbuck, Middleway, Leopards Lane, car park, of the 8.13 Km, 3 km was rehabilitation and 5.13Km was full construction.

The Project value was US $1.4 million. The main works included the civil engineering design, supervision and construction of Bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel wearing course and replenishing or reshaping shoulders and roadside drainage.

Field investigations included Visual survey, traffic survey, topography survey and material survey of construction material.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit. On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase

and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal

Accelerated Urban Roads Rehabilitation Programme 2006/2007 – Solwezi

Solwezi Accelerated Urban Roads Rehabilitation is being carried out on a Design and Build Projects. The Project includes reconstructiing of 5 roads, 7.7km total length. All the selected roads are unsurfaced and will be upgraded to surfaced road.

These being unsurfaced roads, field investigations included In-situ testing (DCP), Laboratory testing of the existing road/pavement material to assess the residual strength. Traffic counts were carried out to ascertain the expected load on the roads.

The recommended interventions after engineering analysis and design were to construct new subbase and base and apply cape seal as surface; other works include replenishing or reshaping shoulders and roadsid

Drainage works to be improved on Kyankwakwa road

e drainage.

Insitu Testing (DCP) of the existing pavement to determine the existing residual strength of the subgrade.

Selected Road before rehabilitation

Accelerated Urban Roads Rehabilitation Programme 2003/2005 – Livingstone

Livingstone Accelerated Urban roads rehabilitation was completed in 2005 December. It was carried out on a design and build basis and the contact value was US $ 1.5million. In order to achieve the objectives of the consultancy, field investigations were carried out with regard to the following aspects:

- Topographical surveys of the roads.
- Evaluation of the existing pavement.
- Survey and analysis of traffic.
- Evaluation of roadside drainage.

These beiing unsurfaced roads, field investigations included In-situ testing (DCP), Laboratory testing of the existing road/pavement material to assess the residual strength. Traffic counts were carried out to ascertain the expected load on the roads.

The recommended interventions after engineering analysis and design were to patch up potholes and seal all cracks then reseal with a Cape seal.

Only two roads required a reconstruction where the existing surface was scarified and compacted as subabse. A new 150mm thick stabilized base was laid and surfaced with a Cape seal. Constructed new subbase and base and apply cape seal as surface,

Other works include replenishing or reshaping shoulders and roadside drainage.

Periodic Maintenance of 102.6 Km of Road (T5) Chingola – Solwezi

Periodic Maintenance of Chingola Solwezi road was funded through the Government of Zambia, Ministry of Works and Supply, Road Fund Agency. The Implementing Agency was the Roads Development Agency.

The length of the road that was worked on was 102.6 km, starting from Chingola on the Copperbelt to Solwezi in North Western Province. Most of the road was in a very bad state, with a lot of pot-holes.

Chingola Solwezi Road before periodic maintenance

Several sections along the road required complete reconstruction; there were a lot of omissions in the original Bill of Quantities (BOQ) that led to variations,

for example, the tonnage of cement, cleaning and de-silting of culverts. Reshaping of side drains to invert

was only billed for the first stretch (km 0 + 000 to Km

36 + 600), there were no quantities for similar works from km 102 + 000 to Km 168 + 000.

The contract included earthworks, drainage works, shoulder repair, road furniture and markings and carriageway reconstruction, including bituminous surface treatments and dressings, broken down as follows:

Earthworks - Reshaping of the shoulder and side drain

Pot hole Patching prior to resurfcaing

Rehabilitation of selected core district and feeder roads in Mumbwa, Kaoma and Mongu Districts: 2005 - 2007

Rehabilitation of District and Feeder Roads in three districts namely Mumbwa, Kaoma and Mongu Districts was funded by the Danish International Development Agency (DANIDA) through Road Development Agency (RDA).

The Project commenced in 2005 and was scheduled to be completed by end of 2007. The Consultancy involved field surveys to identify improvement works required such as earthworks, gravelling requirements and drainage structures.It also included preparation of a confidetial Engineers estimate, Tender Documentation and works supervision.

The Project covered eight 9*No) roads giving a toatal road network of 245.oKm in the three districts.It involved both machine based technology (MBT) and Labour Based Technology (LBT)

The Supervision component involved guiding the contractors in setting out, earthworks, construction of drainage structures, gravelling and quality control.

Accelerated urban roads Rehabilitation programme 2006/2007 – Mansa Town.

Eroded Shoulders on Mwaiseni road

Mansa Accelerated Urban Roads Project was carried out on a Design and Build basis. The Project value was US $1.5million. The Project is expected to finish in June 2006.

The main works included the civil engineering design, supervision and construction of Bituminous surface dressing, rehabilitation and reconstruction of existing pavement or gravel

wearing course and replenishing or reshaping shoulders and roadside drainage.

Field investigation included Visual survey, traffic survey, topography survey and material survey of the existing pavement and prospective borrow pits.

In-situ testing and laboratory testing were carried out on sample of the pavement and the borrow pit.

On new construction, the works were carried by compacting existing base as subgrade, imported gravel for subbase and base. The Base was stabilized with 3% cement before compaction. The surface applied was Cape Seal.

Rehabilitation of Forest Feeder Roads in Luapula and North-Western Province

Rehabilitation of feeder roads in Luapula and North-Western Province was funded by International Fund for Agriculture Development (IFAD)as a sub-component of the Forest Resources Management Project (FRMP). The Project commenced in 2005 and is expected to finish end of 2007.

The works undertaken by the consultant were to assess the roads and design rehabilitation interventions, Cost estimate and drawing preparation, tender document preparation and evaluation the bids.

The Project covered six and five districts in Luapula and North-Western Province respectively. The Project was labour based. The main works carried out were setting out, site clearance, quarrying, earthworks, drainage works, reshaping of the roads to feeder roads standard, gravelling works and other miscellaneous works.

WATER & SANITATION

We specialize in the fields of water and environmental management and engineering and waste management. The company has developed a considerable professional capacity in the management and protection of water resources and in the development of water and wastewater services. The company's activities include:

- Water supply and sanitation strategy development
- Water Supply and Sanitation Engineering
- Water Resources Management
- Waste Management
- Environmental Consulting Digital Map Products and Geographic Information Systems (GIS)

BUILDINGS & INFRASTRUCTURE DEVELOPMENT

We have extensive experience in building works in Zambia and in design procedures and contract administration required by private developers and government departments.

We have undertaken the following projects, in Zambia, in recent years:

- Office Block for Zambia Public Procurement Authority in Lusaka
- Cancer Diseases Hospital Phase II
- Levy Business Park
- Factory for HITACH along Airport Road in Lusaka
- Danish Embassy Car Park
- Chirundu Border Post

Lew Business Park

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